



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2011 HSC Course Assessment Task 1

November 29, 2010

General instructions

- Working time – 50 minutes.
- Commence each new question on a new page.
Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M3A – Mr Lam
- ☐ 12M3B – Mr Weiss
- ☐ 12M3C – Mr Trenwith
- ☐ 12M4A – Mr Fletcher
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mr Rezcallah

NAME: # PAGES USED:

Marker's use only.

QUESTION	1	2	3	4	5	6	Total	%
MARKS	$\overline{9}$	$\overline{9}$	$\overline{9}$	$\overline{7}$	$\overline{10}$	$\overline{9}$	$\overline{53}$	

Question 1 (9 Marks)	Commence a NEW page.	Marks
(a) Given that $\sin A = \frac{2}{3}$ and A is acute, find the value of $\sin 2A$.		2
(b) Find the exact value of $\cos 105^\circ$.		2
(c) Show that $\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$.		2
(d) Solve the equation $2 \sin^2 x = 1 + \cos 2x$ for $0 \leq x \leq 2\pi$.		3

Question 2 (9 Marks)	Commence a NEW page.	Marks
(a) For what value of k does the equation		3
$x^2 + (k - 4)x + 9 = 0$		
have real roots?		
(b) Solve the equation $x^4 = 4(x^2 + 3)$.		3
(c) Write $2x^2 - 3x + 3$ in the form $A(x - 1)^2 + B(x - 1) + C$.		3

Question 3 (9 Marks)	Commence a NEW page.	Marks
(a) A monic polynomial of degree 3 has roots -1 , 1 and 2 . Write the equation of the polynomial in the form		2
$P(x) = ax^3 + bx^2 + cx + d$		
(b) The polynomial $P(x) = px^3 + 5x^2 - 3p$ has $(x - 2)$ as a factor. Find the value of p .		2
(c) Let $f(x) = x^3 + 2x^2 + 5x - 4$.		
i. Show that $f(x)$ has a root between $x = 0$ and $x = 1$.		2
ii. Taking $x = 0.5$ as an approximation to this root, use one application of Newton's method to find a better approximation to the root.		3

Question 4 (7 Marks)	Commence a NEW page.	Marks
(a)	Find the Cartesian equation of the point $P(t + 1, 2t^2 + 1)$.	2
(b)	The curves $y = (x - 1)^2$ and $y = (x + 1)^2$ intersect at the point Q .	
	i. Find the coordinates of Q .	2
	ii. Find the acute angle between the tangents to the curves at Q , giving your answer to the nearest degree.	3

Question 5 (10 Marks)	Commence a NEW page.	Marks
(a)	When the polynomial $P(x)$ is divided by $(x^2 - 1)$, the remainder is $x - 4$. Find the remainder when $P(x)$ is divided by $(x + 1)$.	2
(b)	i. Find the equation of the locus of the point $P(x, y)$ which moves such that its distance from $A(-3, 2)$ is twice its distance from $B(3, -4)$.	3
	ii. Describe this locus geometrically.	2
(c)	One of the roots of $2x^3 + x^2 - 15x - 18 = 0$ is positive and equal to the product of the other two roots.	3

Find all the roots of this equation.

Question 6 (9 Marks)	Commence a NEW page.	Marks
$P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$.		
(a)	Derive the equation of the normal to the parabola at P .	2
(b)	The chord PQ passes through the point $(0, -2)$. Find the equation of the chord PQ and show that $pq = 2$.	3
(c)	The normals at P and Q intersect at R . Show that R lies on the parabola.	4

End of paper.

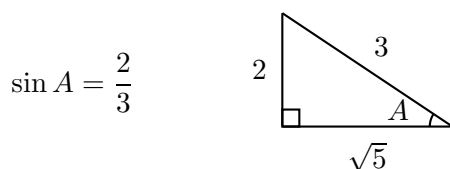
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Suggested Solutions

Question 1 (Fletcher)

(a) (2 marks)

- ✓ [1] for triangle (or equivalent).
- ✓ [1] for final answer in exact form.



$$\sin 2A = 2 \sin A \cos A$$

$$= 2 \times \frac{2}{3} \times \frac{\sqrt{5}}{3} = \frac{4\sqrt{5}}{9}$$

(b) (2 marks)

- ✓ [1] for correct substitution.
- ✓ [1] for final answer.

$$\begin{aligned} \cos 105^\circ &= \cos(60^\circ + 45^\circ) \\ &= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \\ &= \left(\frac{1}{2} \times \frac{1}{\sqrt{2}}\right) - \left(\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}\right) \\ &= \frac{1 - \sqrt{3}}{2\sqrt{2}} \left(= \frac{\sqrt{2} - \sqrt{6}}{4}\right) \end{aligned}$$

(c) (2 marks)

- ✓ [1] for $\frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x}$.
- ✓ [1] for final answer.

$$\begin{aligned} \frac{1 - \tan^2 x}{1 + \tan^2 x} &= \frac{1 - \frac{\sin^2 x}{\cos^2 x} \times \cos^2 x}{1 + \frac{\sin^2 x}{\cos^2 x} \times \cos^2 x} \\ &= \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x} \\ &= \cos 2x \end{aligned}$$

(d) (3 marks)

- ✓ [1] for $4 \sin^2 x = 1$, or equivalent.
- ✓ [1] for $\sin x = \pm \frac{1}{\sqrt{2}}$.
- ✓ [1] for all four solutions. If only $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ is written, a maximum of (2 marks) are awarded.

$$2 \sin^2 x = 1 + \cos 2x$$

$$2 \sin^2 x = 1 + (1 - 2 \sin^2 x)$$

$$4 \sin^2 x = 1$$

$$\sin^2 x = \frac{1}{2} \Rightarrow \sin x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Question 2 (Rezcallah)

(a) (3 marks)

- ✓ [1] for $(k - 4)^2 - 36 \geq 0$.
- ✓ [1] for $(k - 10)(k + 2) \geq 0$.
- ✓ [1] for final answer.

$$x^2 + (k - 4)x + 9 = 0$$

$$\Delta = (k - 4)^2 - 4(9) = (k - 4)^2 - 36$$

There will be real roots when $\Delta \geq 0$:

$$(k - 4)^2 - 36 \geq 0$$

$$(k - 4 - 6)(k - 4 + 6) \geq 0$$

$$(k - 10)(k + 2) \geq 0$$

$$\therefore k \geq 10 \text{ or } k \leq -2$$

(b) (3 marks)

- ✓ [1] for making the substitution and transforming into quadratic (or equivalent).
- ✓ [1] for $x^2 = -2, 6$ (or equivalent).
- ✓ [1] for final answer.

$$x^4 = 4(x^2 + 3)$$

$$x^4 - 4x^2 - 12 = 0$$

Let $m = x^2$,

$$\begin{aligned} m^2 - 4m - 12 &= 0 \\ (m - 6)(m + 2) &= 0 \\ \therefore m &= -2, 6 \\ \therefore x^2 &= -2, 6 \end{aligned}$$

$x^2 = -2$ has no real solutions.

$$\therefore x = \pm\sqrt{6}$$

(c) (3 marks)

- ✓ [1] for each correct value of A , B and C and writing it in the form specified.

$$2x^2 - 3x + 3 \equiv A(x - 1)^2 + B(x - 1) + C \quad \text{(b) (2 marks)}$$

By inspection, $A = 2$. Let $x = 1$,

$$\begin{aligned} 2 - 3 + 3 &= C \\ \therefore C &= 2 \end{aligned}$$

Letting $x = 0$,

$$\begin{aligned} 3 &= 2(-1)^2 + B(-1) + 2 \\ 3 &= 2 - B + 2 \\ -B &= -1 \\ \therefore B &= 1 \\ \therefore 2x^2 - 3x + 3 &\equiv 2(x - 1)^2 + 1(x - 1) + 2 \end{aligned}$$

Question 3 (Lam)

(a) (2 marks)

- ✓ [-1] for each error.

$$P(x) = ax^3 + bx^2 + cx + d$$

- As $P(x)$ is monic, $\therefore a = 1$.
- The roots are $\alpha = -1$, $\beta = 1$ and $\gamma = 2$.
- Apply the sum of roots,

$$\begin{aligned} \alpha + \beta + \gamma &= -\frac{b}{a} \\ -1 + 1 + 2 &= -b \\ \therefore b &= 2 \end{aligned}$$

Apply the sum of pairs of roots,

$$\begin{aligned} \alpha\beta + \beta\gamma + \alpha\gamma &= \frac{c}{a} \\ (-1)(1) + \cancel{(-1)(2)} + \cancel{1(2)} &= c \\ \therefore c &= -1 \end{aligned}$$

Apply the product of roots,

$$\begin{aligned} \alpha\beta\gamma &= -\frac{d}{a} \\ (-1)(1)(2) &= -d \\ \therefore d &= 2 \\ \therefore P(x) &= x^3 - 2x^2 - x + 2 \end{aligned}$$

Alternatively, expand $(x^2 - 1)(x - 2)$.

- ✓ [1] for applying the factor theorem and evaluating $P(2)$.
✓ [1] for final answer.

$$P(x) = px^3 + 5x^2 - 3p$$

By the factor theorem,

$$\begin{aligned} P(2) &= 0 \\ \therefore 2^3p + 5(2^2) - 3p &= 0 \\ 8p + 20 - 3p &= 0 \\ 5p + 20 &= 0 \\ \therefore p &= -4 \end{aligned}$$

(c) i. (2 marks)

- ✓ [1] for evaluating $f(0)$ and $f(1)$.
✓ [1] for final statement (or equivalent).

$$\begin{aligned} f(x) &= x^3 + 2x^2 + 5x - 4 \\ f(0) &= -4 \\ f(1) &= 1 + 2 + 5 - 4 \\ &= 4 \end{aligned}$$

As $f(0) < 0$ and $f(1) > 0$ and f is continuous for all x , therefore $f(x)$ has a root between $x = 0$ and $x = 1$.

ii. (3 marks)

- ✓ [1] for correct differentiation.
- ✓ [2] for final answer. Deduct [1] for each error.

$$\begin{aligned}
 f(x) &= x^3 + 2x^2 + 5x - 4 \\
 f'(x) &= 3x^2 + 4x + 5 \\
 f\left(\frac{1}{2}\right) &= \frac{1}{8} + 2\left(\frac{1}{4}\right) + \frac{5}{2} - 4 = -\frac{7}{8} \\
 f'\left(\frac{1}{2}\right) &= 3\left(\frac{1}{4}\right) + 4\left(\frac{1}{2}\right) + 5 = \frac{31}{4} \\
 x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= \frac{1}{2} - \frac{-\frac{7}{8}}{\frac{31}{4}} \\
 &= \frac{19}{31} \approx 0.6129 \dots
 \end{aligned}$$

Note: $f\left(\frac{19}{31}\right) \approx 0.04 \dots$.
Hence $x = \frac{19}{31} \approx 0.6129$ is a better approximation of the root.

Question 4 (Ireland)

(a) (2 marks)

- ✓ [1] for substituting $t = x - 1$ into $y = 2t^2 + 1$.
- ✓ [1] for final answer $y = 2(x - 1)^2 + 1$.

$$x = t + 1 \quad y = 2t^2 + 1$$

Rearrange $x = t + 1$ and substitute into $y = 2t^2 + 1$:

$$\begin{aligned}
 \therefore y &= 2(x - 1)^2 + 1 \\
 &= 2(x^2 - 2x + 1) + 1 \\
 &= 2x^2 - 4x + 3
 \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for each value of x and y .
- Equate to solve simultaneously:

$$\begin{aligned}
 &\begin{cases} y = (x - 1)^2 \\ y = (x + 1)^2 \end{cases} \\
 &(x + 1)^2 - (x - 1)^2 = 0 \\
 &\left[(x + 1) - (x - 1)\right]\left[(x + 1) + (x - 1)\right] = 0 \\
 &(2)(2x) = 0 \\
 &\therefore x = 0 \\
 &\therefore y = (0 - 1)^2 = 1 \\
 &Q(0, 1)
 \end{aligned}$$

ii. (3 marks)

- ✓ [1] for both values of m_1 and m_2 .
- ✓ [1] for correct substitution into $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$. Max [1] for error in this formula.
- ✓ [1] for final answer.

$$\begin{aligned}
 y &= (x - 1)^2 & \frac{dy}{dx} &= 2(x - 1) \\
 y &= (x + 1)^2 & \frac{dy}{dx} &= 2(x + 1)
 \end{aligned}$$

At $x = 0$, the gradients of the tangents to the curves are

$$m_1 = 2(-1) = -2 \quad m_2 = 2(1) = 2$$

Applying the angle between two lines formula,

$$\begin{aligned}
 \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
 &= \left| \frac{2 - (-2)}{1 + (2)(-2)} \right| = \left| \frac{4}{-3} \right| \\
 \therefore \theta &= 53^\circ \text{ (nearest degree)}
 \end{aligned}$$

Question 5 (Trenwith)

(a) (2 marks)

✓ [1] for rewriting $P(x)$ as the division identity.

✓ [1] for final answer.

$$\begin{aligned} P(x) &= (x^2 - 1)Q(x) + (x - 4) \\ &= (x - 1)(x + 1)Q(x) + (x - 4) \end{aligned}$$

Apply the remainder theorem for division by $x + 1$:

$$P(-1) = 0 + (-1 - 4) = -5$$

Hence the remainder when dividing $P(x)$ by $(x + 1)$ is $R = -5$.

(b) i. (3 marks)

✓ [1] for $PA^2 = 4PB^2$.✓ [1] for substituting expression into $PA^2 = 4PB^2$.✓ [1] for $3x^2 - 30x + 27 + 3y^2 + 36y + 60 = 0$.

$$\begin{aligned} PA &= \sqrt{(x + 3)^2 + (y - 2)^2} \\ PB &= \sqrt{(x - 3)^2 + (y + 4)^2} \\ PA &= 2PB \\ \therefore PA^2 &= 4PB^2 \\ \therefore (x + 3)^2 + (y - 2)^2 &= 4(x - 3)^2 + 4(y + 4)^2 \\ 4(x - 3)^2 + 4(y + 4)^2 &= 0 \\ 4(x^2 - 6x + 9) + 4(y^2 + 8y + 16) &= 0 \\ - (x^2 + 6x + 9) - (y^2 - 4y + 4) &= 0 \\ 3x^2 - 30x + 27 + 3y^2 + 36y + 60 &= 0 \\ 3x^2 - 30x + 3y^2 + 36y + 87 &= 0 \end{aligned}$$

ii. (2 marks)

✓ [1] for circle.

✓ [1] for correct centre and radius.

$$\begin{aligned} \underbrace{3x^2 - 30x + 27 + 3y^2 + 36y + 60}_{\div 3} &= \frac{0}{\div 3} \\ x^2 - 10x + \frac{9}{-9} + y^2 + 12y + \frac{20}{-20} &= \frac{0}{-29} \\ x^2 - 10x + 25 + y^2 + 12y + 36 &= -29 + 25 + 36 \\ (x - 5)^2 + (y + 6)^2 &= 32 \end{aligned}$$

Circle with centre $(5, -6)$ and radius $r = 4\sqrt{2}$.

(c) (3 marks)

✓ [1] for correctly finding the product of roots.

✓ [1] for $2\alpha^2 + 7\alpha + 6 = 0$.

✓ [1] for final answers.

$$2x^3 + x^2 - 15x - 18 = 0$$

Let the roots be α , β and $\alpha\beta$, where $\alpha\beta > 0$. Apply the product of roots,

$$\begin{aligned} \alpha\beta \times \alpha\beta &= \alpha^2\beta^2 = -\frac{d}{a} = 9 \\ \therefore \alpha\beta &= 3 \end{aligned} \quad (5.1)$$

Hence one of the roots is 3. Rearrange,

$$\therefore \alpha = \frac{3}{\beta} \quad (5.2)$$

Apply the sum of roots,

$$\alpha + \beta + \alpha\beta = -\frac{b}{a} = -\frac{1}{2} \quad (5.3)$$

Substitute (5.2) into (5.3),

$$\begin{aligned} \underbrace{\alpha + \frac{3}{\alpha}}_{\times 2\alpha} + 3 &= -\frac{1}{2} \times 2\alpha \\ 2\alpha^2 + 6 + 6\alpha &= -\alpha \\ 2\alpha^2 + 7\alpha + 6 &= 0 \\ (2\alpha + 3)(\alpha + 2) &= 0 \\ \therefore \alpha &= -\frac{3}{2}, -2 \end{aligned}$$

Hence the roots are $-\frac{3}{2}$, -2 and 3 .

Question 6 (Weiss)

(a) (2 marks)

- ✓ [1] for derivation of gradient of normal.
- ✓ [1] for final answer, or equivalent in gradient-intercept form.

$$P(2p, p^2) \quad Q(2q, q^2)$$

Rearrange $x^2 = 4y$,

$$y = \frac{1}{4}x^2 \Rightarrow \frac{dy}{dx} = \frac{1}{2}x$$

At $x = 2p$,

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \times 2p = p \\ \therefore m_{\perp} &= -\frac{1}{p} \end{aligned}$$

Applying the point gradient formula,

$$\begin{aligned} \frac{y - p^2}{x - 2p} &= -\frac{1}{p} \\ py - p^3 &= -x + 2p \\ \therefore x + py &= p^3 + 2p \end{aligned}$$

(b) (3 marks)

- ✓ [1] for correct application of two-point formula.
- ✓ [1] for correct equation of PQ .
- ✓ [1] for correct substitution of $(0, -2)$ into equation of PQ and conclusion.

Apply the two point formula to find the equation of chord PQ :

$$\begin{aligned} \frac{y - q^2}{x - 2q} &= \frac{p^2 - q^2}{2p - 2q} = \frac{(p - q)(p + q)}{2(p - q)} \\ &= \frac{p + q}{2} \end{aligned}$$

$$y - q^2 = \left(\frac{p + q}{2}\right)x - 2q\left(\frac{p + q}{2}\right)$$

$$y - \cancel{q^2} = \left(\frac{p + q}{2}\right)x - pq - \cancel{q^2}$$

$$\therefore y = \left(\frac{p + q}{2}\right)x - pq$$

As the chord PQ passes through $(0, -2)$,

$$\begin{aligned} -2 &= 0 - pq \\ \therefore pq &= 2 \end{aligned}$$

(c) (4 marks)

- ✓ [1] for $y = p^2 + q^2 + pq + 2$.
- ✓ [1] for $x = -2(p + q)$.
- ✓ [1] for use of $pq = 2$ in y coordinate.
- ✓ [1] for correct substitution and conclusion.

From part (a), the normals at P and Q are

$$\begin{cases} x = -py + p^3 + 2p \\ x = -qy + q^3 + 2q \end{cases}$$

Solving simultaneously by equating,

$$\begin{aligned} -py + p^3 + 2p &= -qy + q^3 + 2q \\ py - qy &= p^3 - q^3 + 2p - 2q \\ y(p - q) &= (p^3 - q^3) + 2(p - q) \\ y(\cancel{p - q}) &= (\cancel{p - q})(p^2 + pq + q^2) + 2(\cancel{p - q}) \\ \therefore y &= p^2 + q^2 + 2 + 2 \\ &= p^2 + q^2 + 4 \\ &= p^2 + 2pq + q^2 \\ &= (p + q)^2 \end{aligned}$$

Substitute to find x ,

$$\begin{aligned} x + p(p + q)^2 &= p^3 + 2p \\ x &= -p(p + q)^2 + p^3 + 2p \\ &= -p(p^2 + 2pq + q^2) + p^3 + 2p \\ &= \cancel{p^3} - 2p^2q - pq^2 + \cancel{p^3} + 2p \\ &= -pq(2p + q) + 2p \\ &= -2(2p + q) + 2p \\ &= -4p - 2q + 2p \\ &= -2p - 2q = -2(p + q) \end{aligned}$$

Check that R with coordinates $x = -2(p + q)$ and $y = (p + q)^2$ lies on the parabola $x^2 = 4y$.

$$\begin{aligned} x &= -2(p + q) \\ \therefore p + q &= -\frac{1}{2}x \end{aligned}$$

Substitute to y :

$$\begin{aligned} y &= (p + q)^2 \\ &= \left(-\frac{1}{2}x\right)^2 = \frac{1}{4}x^2 \end{aligned}$$

 $\therefore R$ lies on the parabola as well.